

Genomic DNA extraction (MasterPure Yeast kit)

Adapted from the lab compendium Snowflake Yeast Protocols — Ratcliff Lab / MuLTEE (V2).

Genomic DNA extraction

Genomic DNA is purified using the MasterPure™ Yeast DNA Purification Kit (Epicentre/Lucigen, Cat. Nos. MPY80010/MPY80200).

1. Harvest cells: pellet a saturated 1.5 mL liquid culture by centrifugation ($\geq 10,000$ rpm, 2–5 min), or scrape a single colony (~2 mm diameter) from a plate directly into 300 μ L Yeast Cell Lysis Solution.
2. If starting from a liquid culture pellet, remove all growth medium, then add 300 μ L Yeast Cell Lysis Solution. Optionally add 1 μ L of 5 μ g/ μ L RNase A.
3. Suspend the cells thoroughly by vortexing or by repeated pipetting with a large-bore (1 mL capacity) tip.
4. Incubate at 65 °C for 15 min.
5. Place on ice for 5 min, then add 150 μ L MPC Protein Precipitation Reagent and vortex for 10 s.
6. Pellet cellular debris by centrifugation ($\geq 10,000$ rpm, 10 min).
7. Transfer the supernatant to a clean tube, add 500 μ L isopropanol, and mix by inversion.
8. Pellet the precipitated DNA by centrifugation ($\geq 10,000$ rpm, 10 min).
9. Discard the supernatant; wash the DNA pellet with 0.5 mL 70% ethanol, remove the ethanol, and briefly re-spin to remove any residual liquid.
10. Resuspend the DNA in 35 μ L TE buffer and store at 4 °C.